LEDGER · 2026-08-27 · BIOTECHNOLOGY / BIOCHEMICAL ENGINEERING

Premium Sophorolipids via Solid-State Fermentation of Winterization Oil Cake

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Alkyl Polyglucoside (APG)
Entry application	sulfate-free, green cosmetic cleansers
Measured advantage	none
Time to first revenue	24 months
Gross margin at parity	-198.4%
Startup CapEx	$6,700,000
Capital productivity	0.09x
Regulatory risk	Moderate — qualification costs reported ($1,200,000 estimated)

Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression
- Margin fails the operating-cost check

Assessment

The concept explicitly concedes failure across economic and operational gates, featuring a $6.7M CapEx, 24-month time to first revenue, and a $5.10/kg unit production cost against APG's $1.86/kg market price. The product yields a sticky crude paste requiring localized heating/solvents, representing a severe form factor regression from liquid APG. Additionally, the sophorolipids suffer from salt intolerance and acidic aggregation in formulation.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts